In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df = pd.read_csv("C:/Users/chait/Downloads/googleplaystore.csv")
```

In [3]:
```
df.head()
```

Out[3]:

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Cont Rat
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Every
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Every
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Every
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25M	50,000,000+	Free	0	T
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2.8M	100,000+	Free	0	Every

In [4]:
```
df.describe()
```

Out[4]:

	Rating
count	9367.000000
mean	4.193338
std	0.537431
min	1.000000
25%	4.000000
50%	4.300000
75%	4.500000
max	19.000000

In [5]:
```
df.dtypes
```

Out[5]:
```
App                object
Category           object
Rating            float64
Reviews            object
Size               object
Installs           object
Type               object
Price              object
Content Rating     object
Genres             object
Last Updated       object
Current Ver        object
Android Ver        object
dtype: object
```

In [6]:
```
df.columns
```

Out[6]:
```
Index(['App', 'Category', 'Rating', 'Reviews', 'Size', 'Installs', 'Type',
       'Price', 'Content Rating', 'Genres', 'Last Updated', 'Current Ver',
       'Android Ver'],
      dtype='object')
```

In [7]:
```
df.duplicated().sum()
```

Out[7]:
```
np.int64(483)
```

In [8]:
```
df.drop_duplicates(inplace = True)
```

In [9]:
```
df.shape
```

Out[9]:
```
(10358, 13)
```

In [10]:
```
df.duplicated().sum()
```

Out[10]:
```
np.int64(0)
```

In [11]:
```
df.isnull().sum()
```

Out[11]:
```
App                  0
Category             0
Rating            1465
Reviews              0
Size                 0
Installs             0
Type                 1
Price                0
Content Rating       1
Genres               0
Last Updated         0
Current Ver          8
Android Ver          3
dtype: int64
```

In [12]:
```
df['Rating'].unique()
```

Out[12]:
```
array([ 4.1,  3.9,  4.7,  4.5,  4.3,  4.4,  3.8,  4.2,  4.6,  3.2,  4. ,
        nan,  4.8,  4.9,  3.6,  3.7,  3.3,  3.4,  3.5,  3.1,  5. ,  2.6,
        3. ,  1.9,  2.5,  2.8,  2.7,  1. ,  2.9,  2.3,  2.2,  1.7,  2. ,
        1.8,  2.4,  1.6,  2.1,  1.4,  1.5,  1.2, 19. ])
```

In [13]:
```
df['Rating'] = df['Rating'].fillna(df['Rating'].mean())
```

In [14]:
```
df.isnull().sum()
```

Out[14]:
```
App               0
Category          0
Rating            0
Reviews           0
Size              0
Installs          0
Type              1
Price             0
Content Rating    1
Genres            0
Last Updated      0
Current Ver       8
Android Ver       3
dtype: int64
```

In [15]:
```
df['Type'] = df['Type'].fillna(df['Type'].mode()[0])
```

In [16]:
```
df.isnull().sum()
```

Out[16]:
```
App               0
Category          0
Rating            0
Reviews           0
Size              0
Installs          0
Type              0
Price             0
Content Rating    1
Genres            0
Last Updated      0
Current Ver       8
Android Ver       3
dtype: int64
```

In [17]:
```
df['Content Rating'].unique()
```

Out[17]:
```
array(['Everyone', 'Teen', 'Everyone 10+', 'Mature 17+',
       'Adults only 18+', 'Unrated', nan], dtype=object)
```

In [18]:
```
df.dropna(subset = ['Content Rating'] , inplace = True)
```

In [19]:
```
df.isnull().sum()
```

Out[19]:
```
App                0
Category           0
Rating             0
Reviews            0
Size               0
Installs           0
Type               0
Price              0
Content Rating     0
Genres             0
Last Updated       0
Current Ver        8
Android Ver        2
dtype: int64
```

In [20]:
```
df.shape
```

Out[20]:
```
(10357, 13)
```

In [21]:
```
df['Current Ver']
```

Out[21]:
```
0                      1.0.0
1                      2.0.0
2                      1.2.4
3         Varies with device
4                        1.1
                ...
10836                   1.48
10837                    1.0
10838                    1.0
10839     Varies with device
10840     Varies with device
Name: Current Ver, Length: 10357, dtype: object
```

In [22]:
```
df['Current Ver'] = df['Current Ver'].replace('Varies with device',np.nan)
```

In [23]:
```
df['Current Ver'].isnull().sum()
```

Out[23]:
```
np.int64(1310)
```

In [24]:
```
df['currnt Ver'] = df['Current Ver'].fillna('Unknown')
```

In [25]:
```
df.isnull().sum()
```

Out[25]:
```
App                  0
Category             0
Rating               0
Reviews              0
Size                 0
Installs             0
Type                 0
Price                0
Content Rating       0
Genres               0
Last Updated         0
Current Ver       1310
Android Ver          2
currnt Ver           0
dtype: int64
```

In [26]:
```
 df.dropna( subset = ['Android Ver'] , inplace =True)
```

In [27]:
```
df.isnull().sum()
```

Out[27]:
```
App                  0
Category             0
Rating               0
Reviews              0
Size                 0
Installs             0
Type                 0
Price                0
Content Rating       0
Genres               0
Last Updated         0
Current Ver       1310
Android Ver          0
currnt Ver           0
dtype: int64
```

In [28]:
```
df['Android Ver'].unique()
```

Out[28]:
```
array(['4.0.3 and up', '4.2 and up', '4.4 and up', '2.3 and up',
       '3.0 and up', '4.1 and up', '4.0 and up', '2.3.3 and up',
       'Varies with device', '2.2 and up', '5.0 and up', '6.0 and up',
       '1.6 and up', '1.5 and up', '2.1 and up', '7.0 and up',
       '5.1 and up', '4.3 and up', '4.0.3 - 7.1.1', '2.0 and up',
       '3.2 and up', '4.4W and up', '7.1 and up', '7.0 - 7.1.1',
       '8.0 and up', '5.0 - 8.0', '3.1 and up', '2.0.1 and up',
       '4.1 - 7.1.1', '5.0 - 6.0', '1.0 and up', '2.2 - 7.1.1',
       '5.0 - 7.1.1'], dtype=object)
```

In [29]:
```
df['Android Ver'] = df['Android Ver'].replace('Varies with device',np.nan)
```

In [30]:
```
df['Android Ver'].isnull().sum()
```

Out[30]:
```
np.int64(1221)
```

In [31]:
```
df['Android Ver'] = df['Android Ver'].fillna(df['Android Ver'].mode()[0])
```

In [32]:
```
df.isnull().sum()
```

Out[32]:
```
App                 0
Category            0
Rating              0
Reviews             0
Size                0
Installs            0
Type                0
Price               0
Content Rating      0
Genres              0
Last Updated        0
Current Ver      1310
Android Ver         0
currnt Ver          0
dtype: int64
```

In [33]:
```
df.head()
```

Out[33]:

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Cont Rat
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Every
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Every
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Every
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25M	50,000,000+	Free	0	T
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2.8M	100,000+	Free	0	Every

In [34]:
```
df.columns
```

Out[34]:
```
Index(['App', 'Category', 'Rating', 'Reviews', 'Size', 'Installs', 'Type',
       'Price', 'Content Rating', 'Genres', 'Last Updated', 'Current Ver',
       'Android Ver', 'currnt Ver'],
      dtype='object')
```

In [35]:
```
df = df.drop(['Current Ver'] , axis = 1)
```

In [36]:
```
df.head()
```

Out[36]:

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Cont Rat
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Every
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Every
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Every
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25M	50,000,000+	Free	0	T
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2.8M	100,000+	Free	0	Every

In [37]:
```
df.Price.unique()
```

Out[37]:
```
array(['0', '$4.99', '$3.99', '$6.99', '$1.49', '$2.99', '$7.99', '$5.99',
       '$3.49', '$1.99', '$9.99', '$7.49', '$0.99', '$9.00', '$5.49',
       '$10.00', '$24.99', '$11.99', '$79.99', '$16.99', '$14.99',
       '$1.00', '$29.99', '$12.99', '$2.49', '$10.99', '$1.50', '$19.99',
       '$15.99', '$33.99', '$74.99', '$39.99', '$3.95', '$4.49', '$1.70',
       '$8.99', '$2.00', '$3.88', '$25.99', '$399.99', '$17.99',
       '$400.00', '$3.02', '$1.76', '$4.84', '$4.77', '$1.61', '$2.50',
       '$1.59', '$6.49', '$1.29', '$5.00', '$13.99', '$299.99', '$379.99',
       '$37.99', '$18.99', '$389.99', '$19.90', '$8.49', '$1.75',
       '$14.00', '$4.85', '$46.99', '$109.99', '$154.99', '$3.08',
       '$2.59', '$4.80', '$1.96', '$19.40', '$3.90', '$4.59', '$15.46',
       '$3.04', '$4.29', '$2.60', '$3.28', '$4.60', '$28.99', '$2.95',
       '$2.90', '$1.97', '$200.00', '$89.99', '$2.56', '$30.99', '$3.61',
       '$394.99', '$1.26', '$1.20', '$1.04'], dtype=object)
```

In [38]:
```
df['Price'] = df['Price'].str.replace("$","", regex = False)
```

In [39]:
```
df['Price'].unique()
```

```
Out[39]: array(['0', '4.99', '3.99', '6.99', '1.49', '2.99', '7.99', '5.99',
                '3.49', '1.99', '9.99', '7.49', '0.99', '9.00', '5.49', '10.00',
                '24.99', '11.99', '79.99', '16.99', '14.99', '1.00', '29.99',
                '12.99', '2.49', '10.99', '1.50', '19.99', '15.99', '33.99',
                '74.99', '39.99', '3.95', '4.49', '1.70', '8.99', '2.00', '3.88',
                '25.99', '399.99', '17.99', '400.00', '3.02', '1.76', '4.84',
                '4.77', '1.61', '2.50', '1.59', '6.49', '1.29', '5.00', '13.99',
                '299.99', '379.99', '37.99', '18.99', '389.99', '19.90', '8.49',
                '1.75', '14.00', '4.85', '46.99', '109.99', '154.99', '3.08',
                '2.59', '4.80', '1.96', '19.40', '3.90', '4.59', '15.46', '3.04',
                '4.29', '2.60', '3.28', '4.60', '28.99', '2.95', '2.90', '1.97',
                '200.00', '89.99', '2.56', '30.99', '3.61', '394.99', '1.26',
                '1.20', '1.04'], dtype=object)
```

```
In [40]: df['Price'] = df['Price'].astype(float)
```

```
In [41]: df.dtypes
```

```
Out[41]: App               object
         Category          object
         Rating           float64
         Reviews           object
         Size              object
         Installs          object
         Type              object
         Price            float64
         Content Rating    object
         Genres            object
         Last Updated      object
         Android Ver       object
         currnt Ver        object
         dtype: object
```

```
In [42]: df['Installs'] = (df['Installs'].str.replace("+","", regex = False )
                                          .str.replace("," , "" , regex = False))
```

```
In [43]: df['Installs'] = df['Installs'].astype(int)
```

```
In [44]: df.dtypes
```

```
Out[44]: App               object
         Category          object
         Rating           float64
         Reviews           object
         Size              object
         Installs           int64
         Type              object
         Price            float64
         Content Rating    object
         Genres            object
         Last Updated      object
         Android Ver       object
         currnt Ver        object
         dtype: object
```

```
In [45]: df['Reviews'] = pd.to_numeric(df['Reviews'] , errors = 'coerce')
```

```
In [46]: df['Reviews'].isnull().sum()
```

Out[46]: np.int64(0)

In [47]:
```
df.dtypes
```

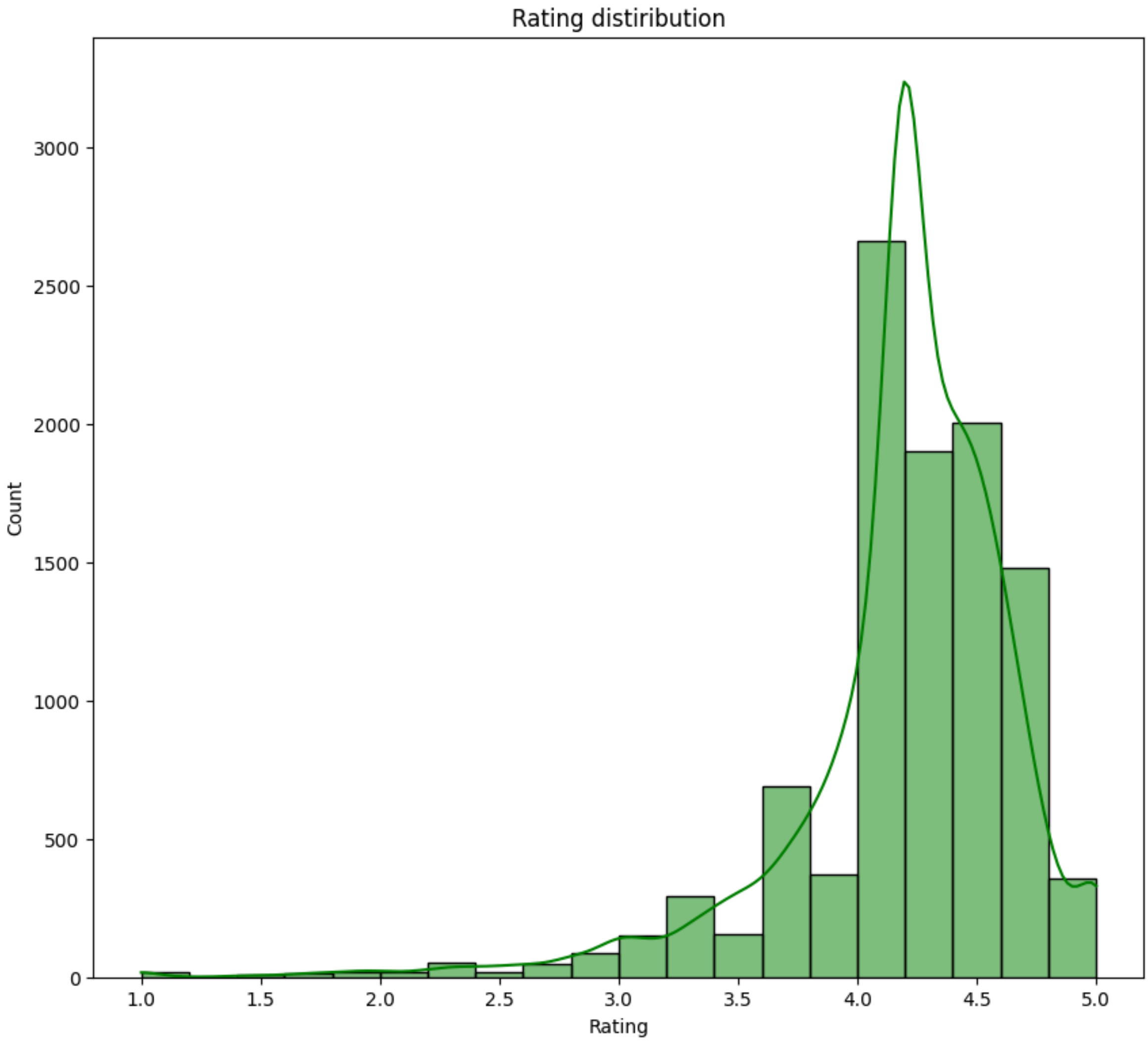
Out[47]:
```
App                 object
Category            object
Rating             float64
Reviews              int64
Size                object
Installs             int64
Type                object
Price              float64
Content Rating      object
Genres              object
Last Updated        object
Android Ver         object
currnt Ver          object
dtype: object
```

In [48]:
```
df['Last Updated'] = pd.to_datetime(df['Last Updated'])
```

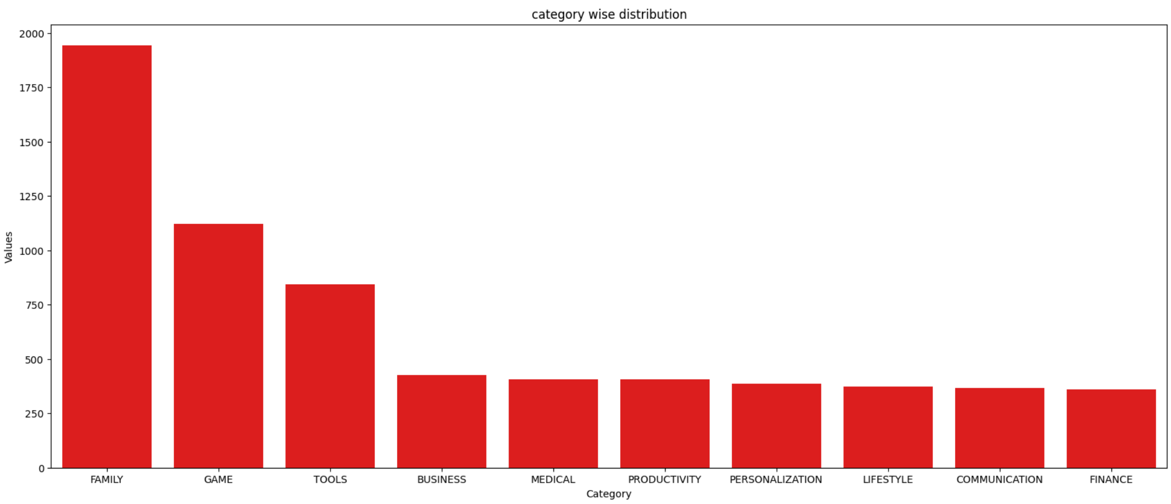
In [49]:
```
df.dtypes
```

Out[49]:
```
App                        object
Category                   object
Rating                    float64
Reviews                     int64
Size                       object
Installs                    int64
Type                       object
Price                     float64
Content Rating             object
Genres                     object
Last Updated       datetime64[ns]
Android Ver                object
currnt Ver                 object
dtype: object
```

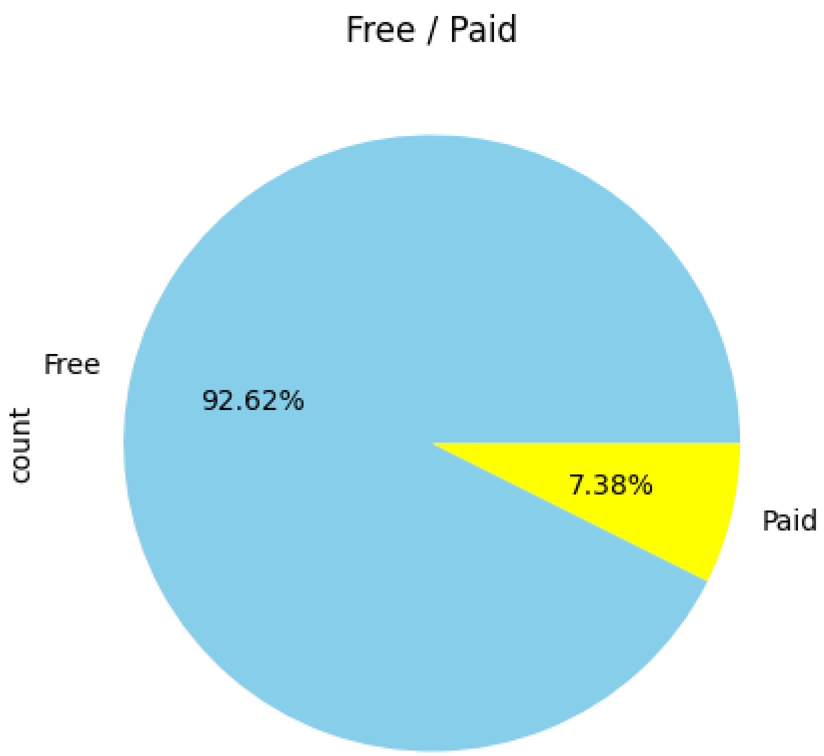
In [50]:
```
plt.figure( figsize = (10,9))
sns.histplot(df['Rating'] , bins = 20 , kde = True , color = "Green" )
plt.title("Rating distiribution")
plt.show()
```



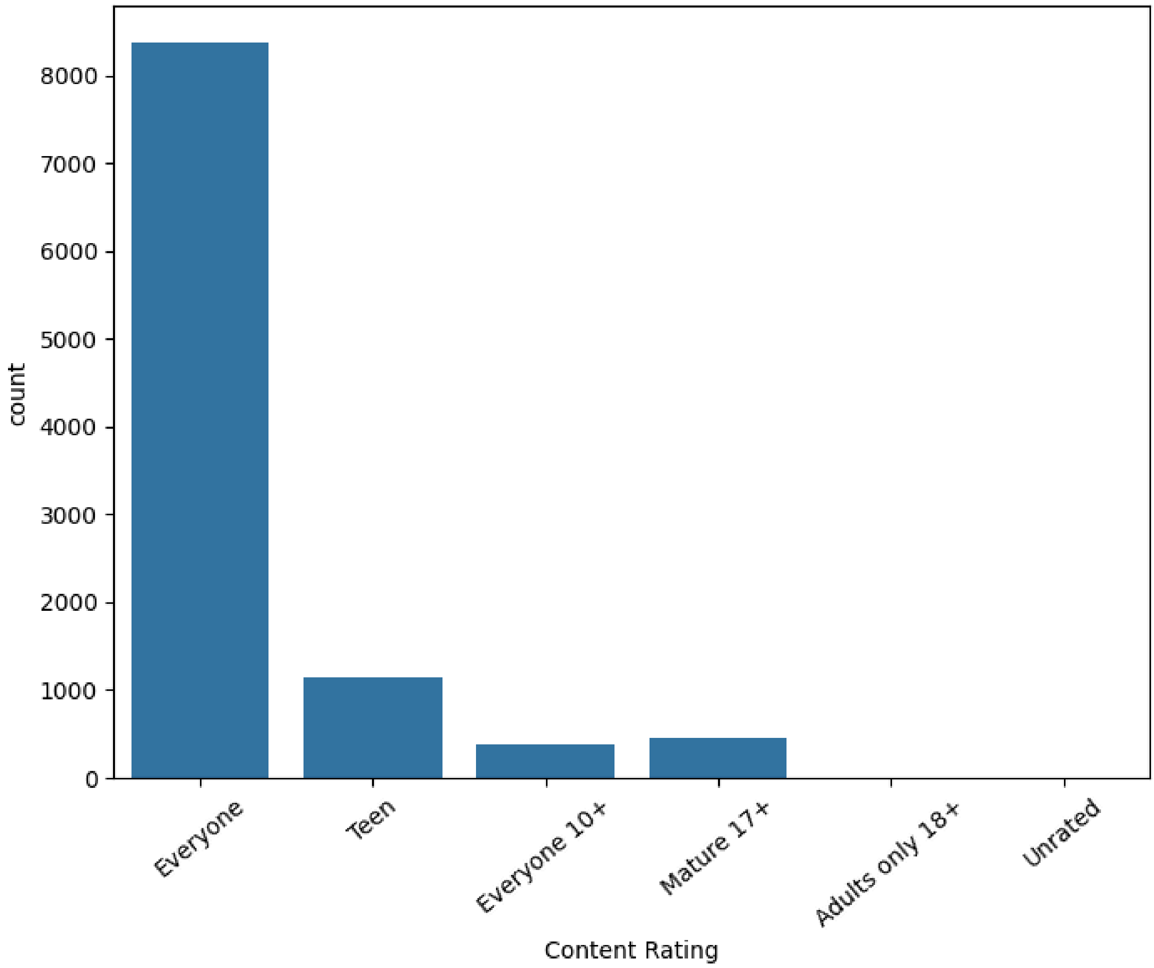
In [51]:
```
plt.figure(figsize = (20,8))
top_categories = df['Category'].value_counts().head(10)
sns.barplot(y=top_categories.values , x= top_categories.index , color = 'Red')
plt.xlabel('Category')
plt.ylabel('Values')
plt.title("category wise distribution")
plt.show()
```



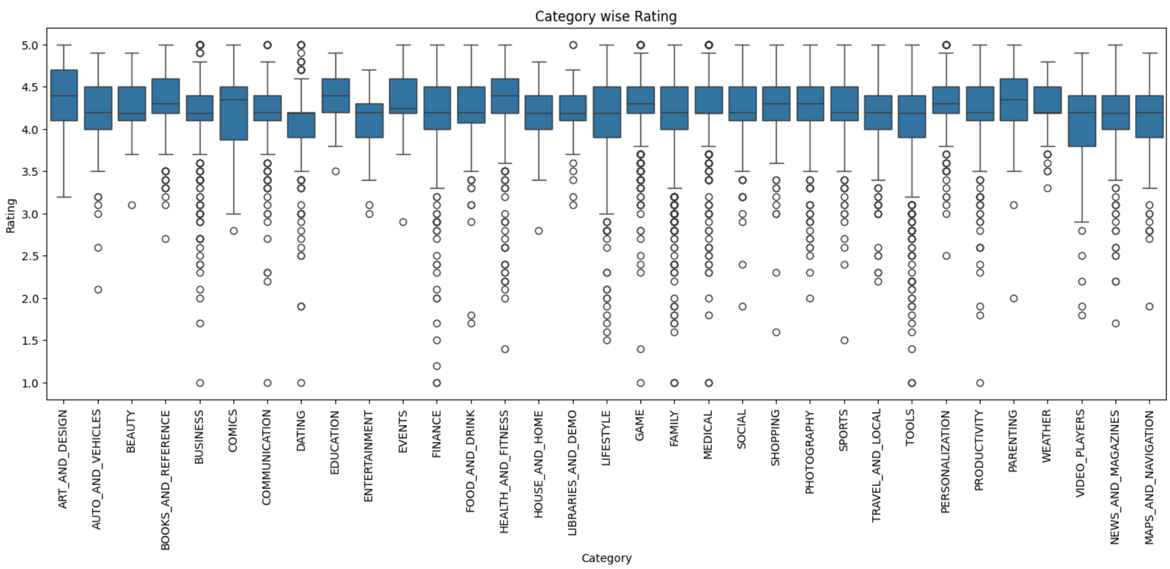
In [52]:
```
df['Type'].value_counts().plot(kind = "pie" , autopct = "%1.2f%%" , figsize = (5
plt.title("Free / Paid")
plt.show()
```



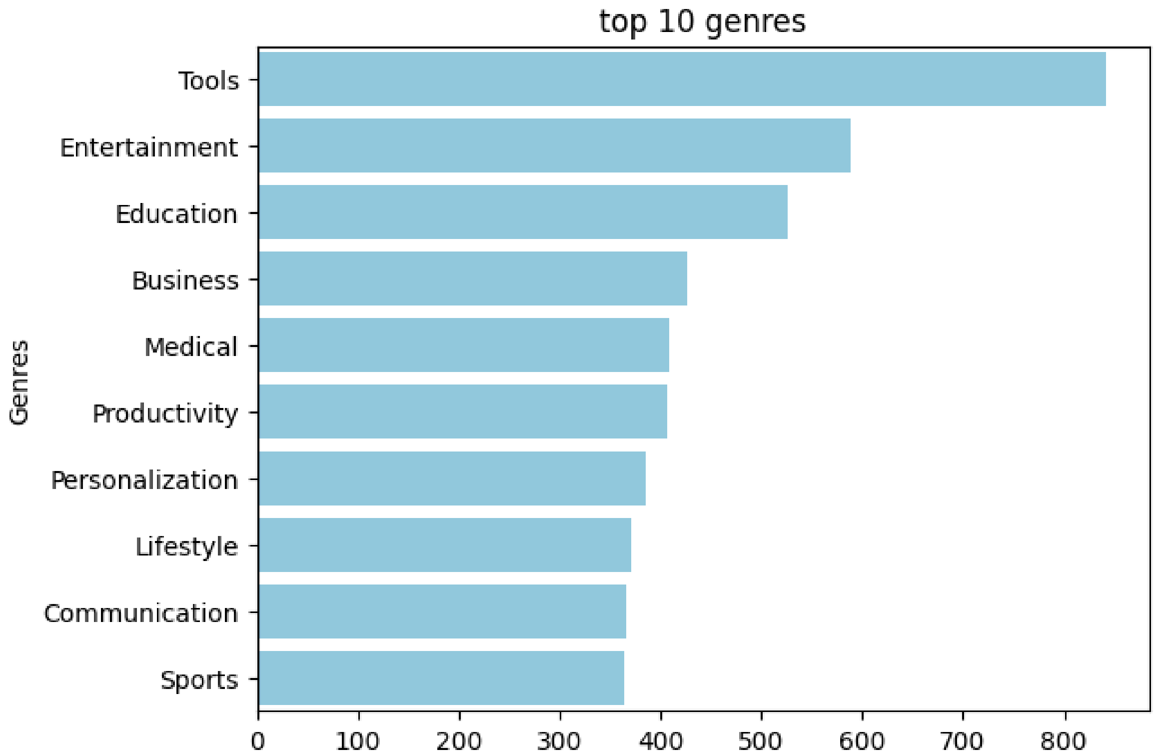
In [53]:
```
plt.figure( figsize = (8,6))
sns.countplot( data = df , x = 'Content Rating')
plt.xticks(rotation = 40)
plt.show()
```



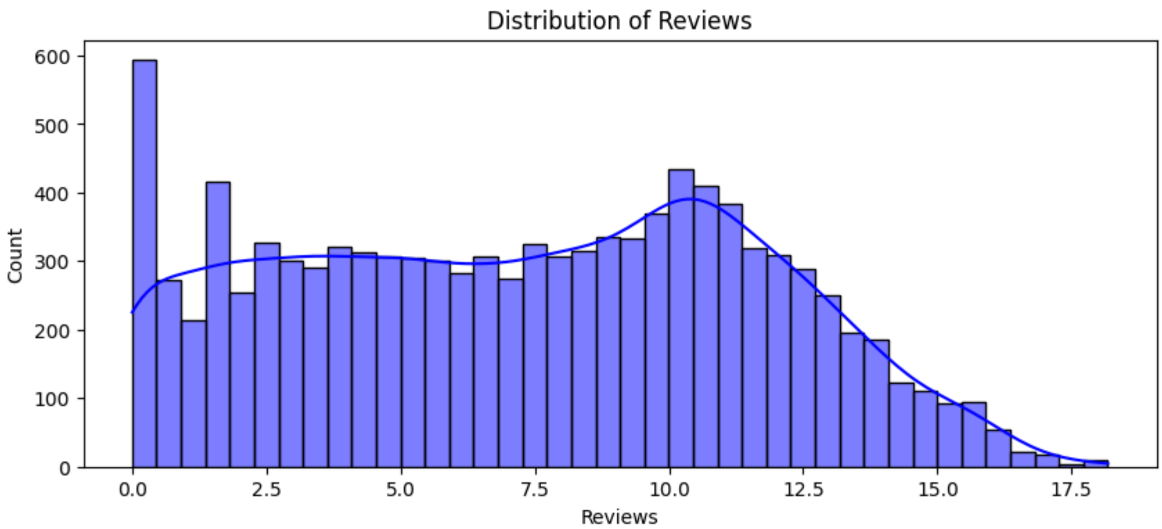
In [54]:
```
plt.figure(figsize = (18,6))
sns.boxplot(data = df , x = 'Category' , y = 'Rating')
plt.xticks( rotation = 90)
plt.title("Category wise Rating")
plt.show()
```



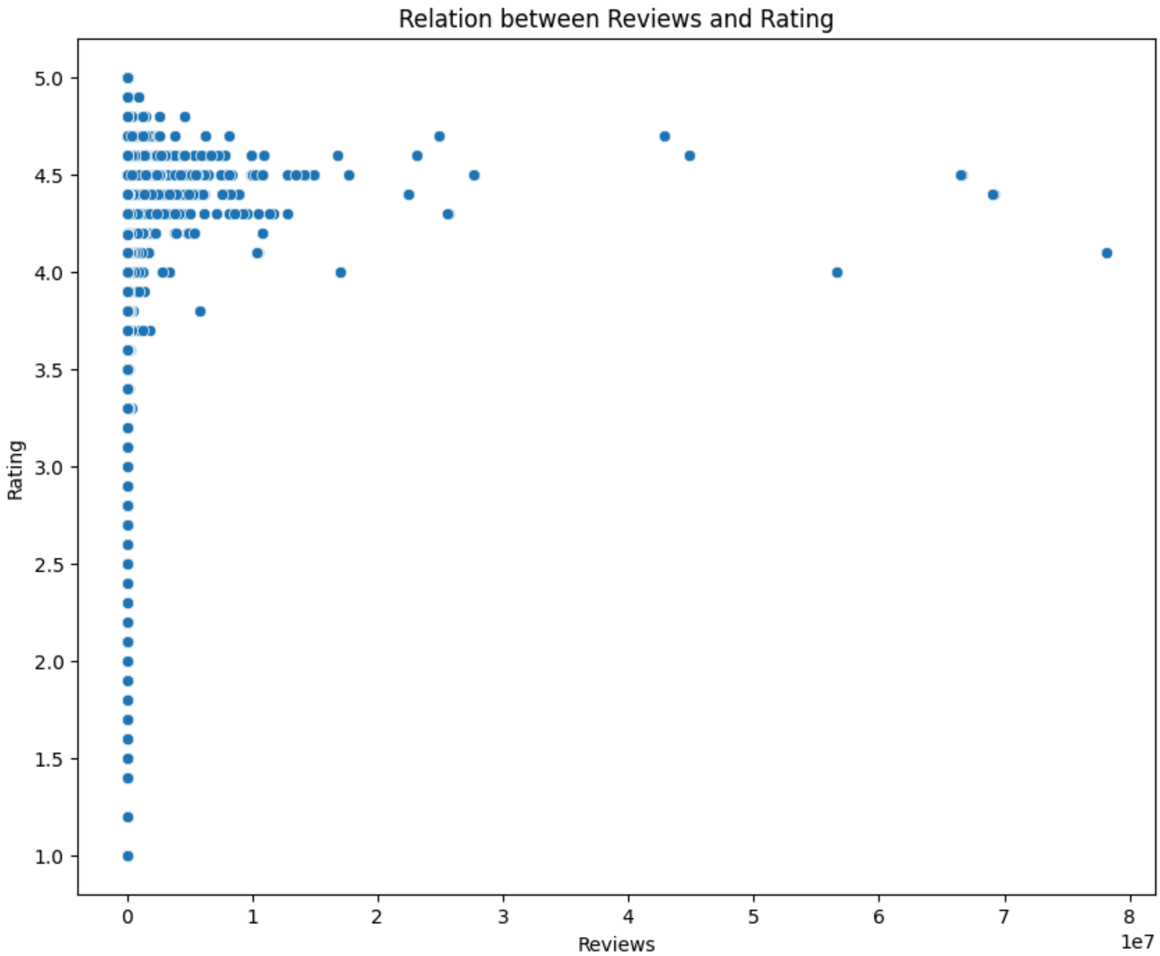
In [55]:
```
top_genre = df['Genres'].value_counts().head(10)
sns.barplot(y = top_genre.index , x = top_genre.values , color = "Skyblue")
plt.title("top 10 genres")
plt.show()
```



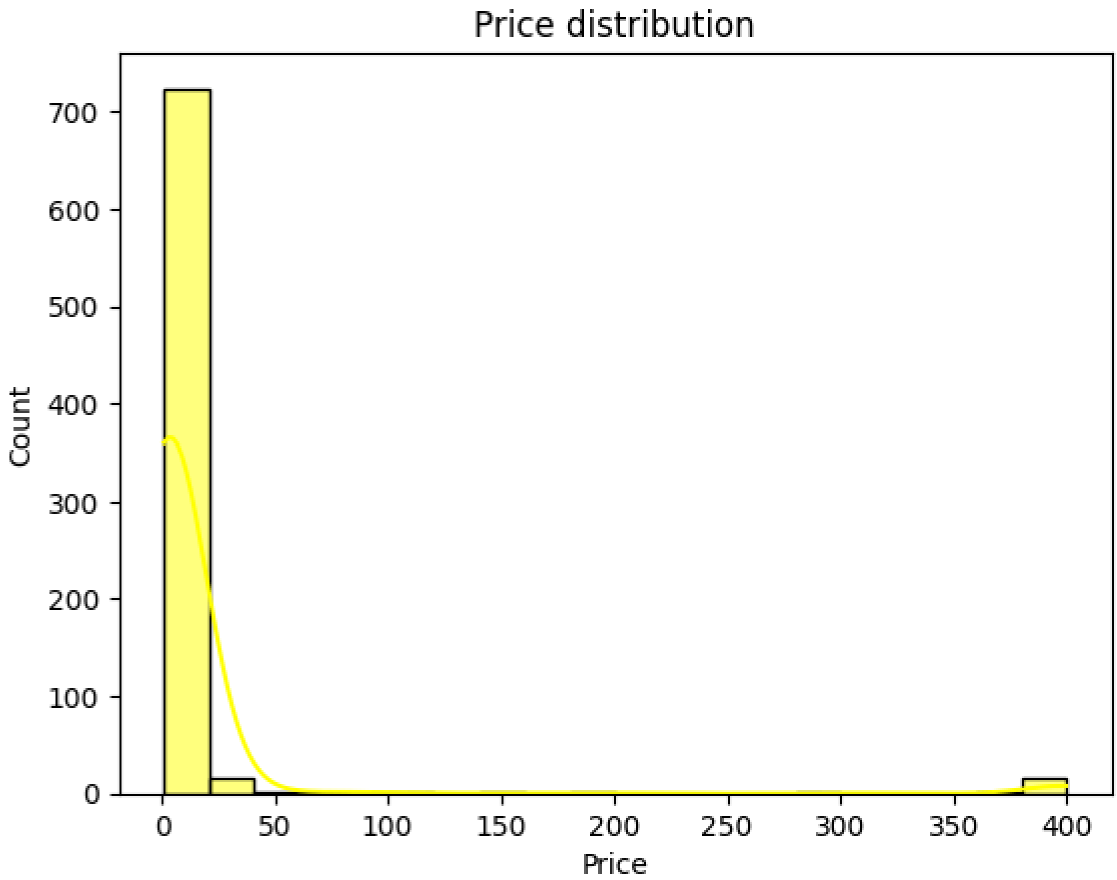
In [60]:
```
plt.figure(figsize = (10,4))
sns.histplot(np.log1p(df['Reviews']) ,bins = 40 , color = 'blue' ,kde = True)
plt.title("Distribution of Reviews")
plt.show()
```



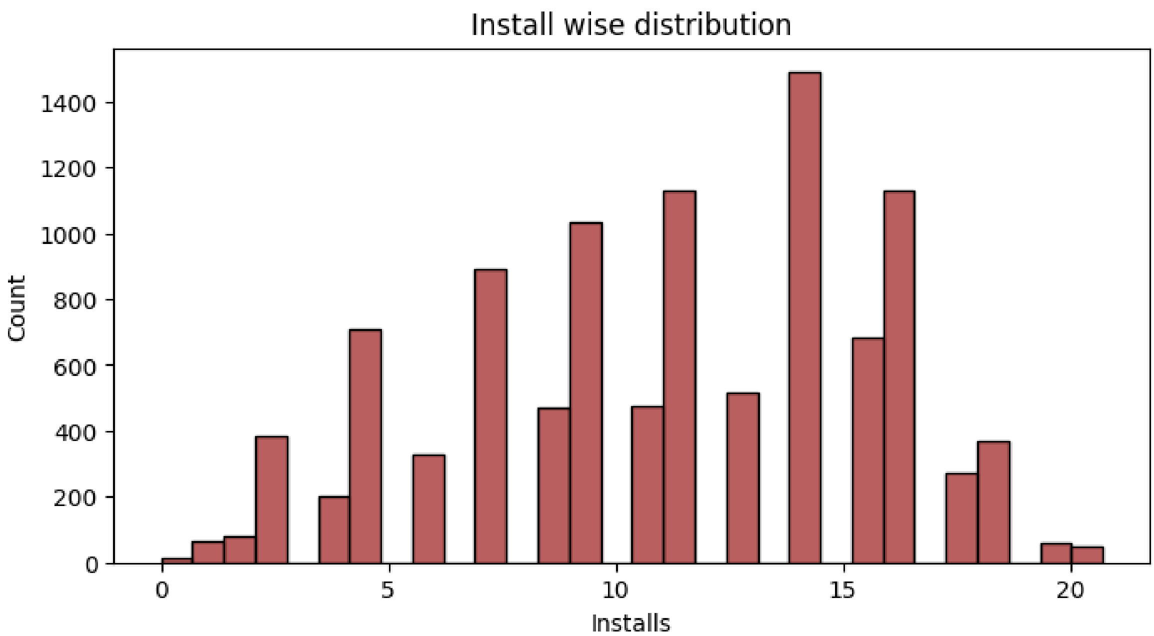
In [62]:
```
plt.figure(figsize = (10,8))
sns.scatterplot( data = df , x = 'Reviews' , y = 'Rating' )
plt.title('Relation between Reviews and Rating')
plt.xlabel('Reviews')
plt.ylabel('Rating')
plt.show()
```



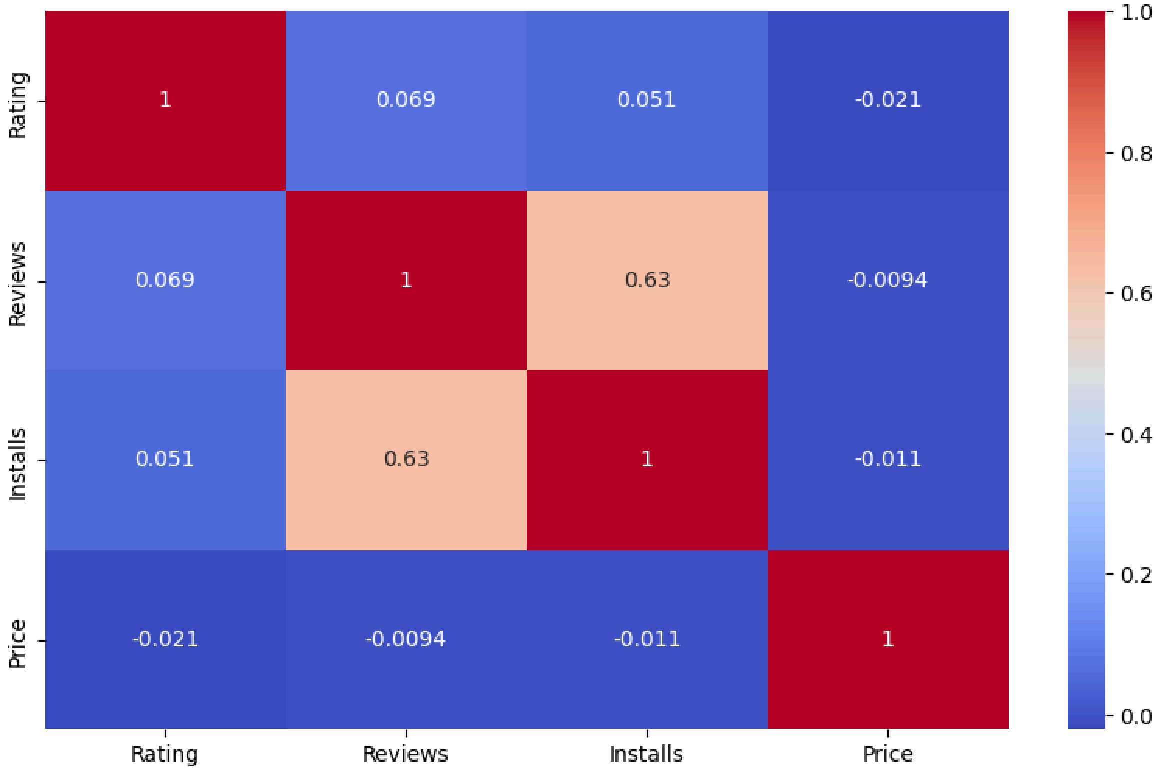
In [63]:
```
paid = df[df['Type']=='Paid']
sns.histplot(paid['Price'] , color = 'Yellow' , kde = True , bins = 20)
plt.title('Price distribution')
plt.show()
```



In [67]:
```
plt.figure(figsize = (8,4))
sns.histplot(np.log1p(df['Installs'] ), bins = 30 , color = 'brown' )
plt.title('Install wise distribution')
plt.show()
```



In [71]:
```
am = df[["Rating","Reviews","Installs","Price"]]
plt.figure(figsize = (10,6))
sns.heatmap(am.corr() , annot =True ,cmap = 'coolwarm')
plt.show()
```

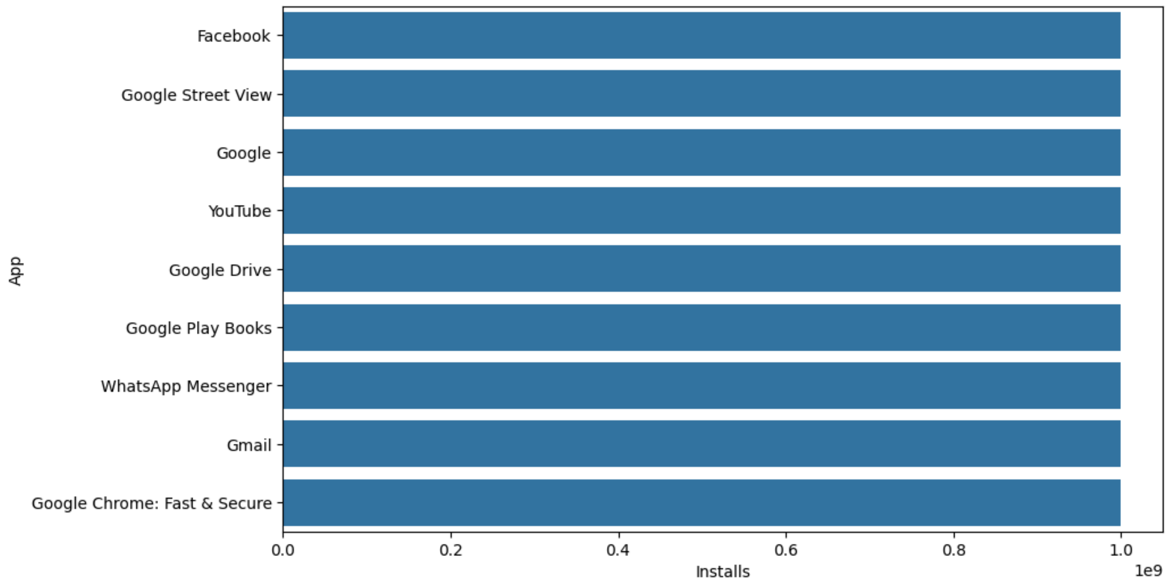


In [75]:
```
top_apps = df.sort_values(
by="Installs",
ascending=False
).head(10)

plt.figure(figsize=(10,6))

sns.barplot(
x="Installs",
y="App",
data=top_apps
)

plt.show()
```



In [ ]:
```
- Most playstore app are free
- Average rating lie between the 4 and 4.5
```

```
- Family category have the highest number of app
- Tools and Entertainment have the highest count among the genres
- There is no relation between the Rating and Reviews
- most of app price are under the $10
- There is strong relation between the Reviews and Installs
- Categories such as Game, Tools, Productivity, and Communication have high user
```